Benchmarking Channel Pruning for Few-Shot Earth Observation

Anonymous WACV Applications Track submission

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Abstract

Which channels should be kept when a pretrained network is pruned to recognise classes it was never trained on, from a handful of labelled examples? We compare eight channel scores on a ResNet-18 and five of them on a MobileNetV2: random choice, filter ℓ_1 norm, BatchNorm scale, a first-order Taylor term, DACIS (a rank sum of three statistics) and each of those statistics alone. Test classes are held out from NWPU-RESISC45 aerial scenes and 13-band Sentinel-2 EuroSAT-MS patches, with leaf photographs as a ground-level control; five to ten random class partitions, one set of test episodes for all methods. DACIS is never reliably ahead of random choice. The Taylor term is ahead of it by 1.4 points on RESISC45 for the ResNet-18 at 30% of the parameters, and by 6.0 and 6.5 points at the severest budgets we tried (ResNet-18 at 10%, MobileNetV2 at 25%), where BatchNorm scale on the MobileNetV2 falls 8.3 points below random choice. The meta-learning stage of a prune, meta-train, prune schedule did not raise 5-shot accuracy, and a naive 13-band input did not beat RGB. Pruning the ResNet-18 to 30% removed 58% of the multiply-accumulates but only 39% of CPU latency, and an unpruned MobileNetV3-Small, at 0.018 G multiply-accumulates, beats every pruned network we built on the three novel-class sets. Notebooks rerun the experiments and rebuild every table.

1. Introduction

Earth-observation payloads and field units must recognise new scene types from a few labelled examples and with little compute. Take five labelled aerial scenes for each of five classes held out from training. A ResNet-18 trained on 25 other NWPU-RESISC45 classes picks the right one of the five for 87.2% of the query scenes (chance is 20%). Keep 30% of its parameters and the figure falls to 81.0% with the DACIS score, to 81.1% with randomly chosen channels and to 82.4% with a first-order Taylor score. Are such differences real, and what else moves the number?

We care about small hardware that cannot be retrained often, such as the payload of a small satellite. On-board inference already flies [11, 12], the classes that matter change with the mission, and labels for a new class are few. Pruning is one route to a smaller network, and a prototype head can then be updated from five labelled patches without touching the weights; our cost results question whether it is the best route. The pruning studies we build on [15, 20, 23, 27] report accuracy on the classes the network was trained for. What survives when the classes at test time are not those the channels were chosen for?

We study that on a ResNet-18 [14] and on a MobileNetV2 [31]. Every method removes inner channels of the residual or inverted-residual blocks under one shared depth profile (Sec. 3), so two scores differ only in which channels they keep. The scores are random choice, filter ℓ_1 norm, BatchNorm scale, a Taylor term, and DACIS, a rank sum of gradient norm, activation variance and Fisher ratio over class labels, which we also run in its three parts. Among the schedules is PMP: prune, meta-train, prune again. Everything is tested on held-out classes, over many class partitions and with shared test episodes. The test sets are aerial scenes (NWPU-RESISC45 [5]) and Sentinel-2 patches with all thirteen bands (EuroSAT-MS [16]). PlantVillage leaf photographs [19], with PlantDoc field photographs [32] as a shift check, are a ground-level control. They lack geospatial context, but the episodic learning rate was chosen on PlantVillage and the DACIS ablations run on it (Secs. 4 and 5).

Contributions. (i) A protocol for pruning studies in few-shot Earth observation: classes are held out, partitions are replicated and all methods share test episodes, so paired intervals can be reported. (ii) To our knowledge the first comparison of channel scores on held-out classes: eight scores on a ResNet-18 and five on a Mo-

bileNetV2, on RESISC45 and thirteen-band EuroSAT-MS. (iii) A cost study and a compact-network baseline. Findings:

- Only Taylor beats random choice. On the ResNet-18 at 30% of the parameters DACIS cannot be told from random choice; Taylor is ahead by 1.4 points on RESISC45, a lead that survives correction for the 28 PMP comparisons we made.
- The score matters more under severe pruning. At 10% of the ResNet-18's parameters and 25% of the MobileNetV2's, Taylor is ahead of random choice on RESISC45 by 6.0 and 6.5 points, and BatchNorm scale is 8.3 points behind on the MobileNetV2.
- Meta-training does not help. Episodic training does not raise 5-shot accuracy; the few schedules that gain over plain prune-and-recover are those whose score uses the meta-gradient statistic, and a variant that updates no weight gains as much.
- A small pretrained network wins. Pruning saves less time than operations (58% fewer multiply-accumulates, 39% lower CPU latency), and an unpruned MobileNetV3-Small beats every pruned network we built on the three novel-class sets. Pruning studies should include it as a baseline.

Sec. 6 lists what we did not test: coordinates, dates, radar, full scenes, satellite-pretrained backbones and embedded hardware. This is a benchmarking study on Earth-observation data, not a geospatial method.

2. Related work

Channel pruning. Most pipelines start from one of three scores: the ℓ_1 norm of a filter [20], the BatchNorm scale learned under a sparsity penalty [23], or a first-order Taylor estimate of the loss change [27, 28]. He et al. [15] choose channels that best reconstruct the next layer's output and, for residual networks, prune only the inner convolutions of each block, which is the scheme we use. Discrimination-aware pruning [38] adds auxiliary classification losses to intermediate layers and keeps the channels that preserve them. DACIS asks a similar question with a cheaper statistic, the Fisher ratio of the pooled activation, and no extra training. These methods are evaluated on the classes they were trained for. Two earlier results suggest that such an evaluation tells us less than it seems to. Mittal et al. [26] showed that removing 25 to 50% of the filters of VGG-16 and ResNet-50 at random and then fine-tuning matches ranked pruning, so that recovery, not the ranking, carries the result. Liu et al. [25] found that a fine-tuned pruned network is no better than the same architecture trained from scratch. MetaPruning [24] also says "meta", but there a hypernetwork predicts weights for candidate architectures; our meta stage is ordinary episodic training.

Is the ranking needed? Random choice has been strong in other pruning settings. Su et al. [34] and Li et al. [22] report that randomly pruned networks with suitable layer-wise widths are competitive with criterion-based pruning, and Blalock et al. [2] document how often pruning papers compare against weak baselines or inconsistently with one another. All of these evaluate on the classes the network was trained for. Tian et al. [36] prune the network inside a meta-learner to reduce overfitting to the training tasks, but compare no channel scores. We did not find a comparison of channel scores on classes held out from the ones used to choose the channels. This paper makes one, with paired test episodes over several class partitions.

Few-shot classification. Prototypical networks [33] label a query by the nearest class mean in embedding space, and MAML [9] learns an initialisation that adapts in a few gradient steps. A network pre-trained with plain cross-entropy on the base classes is a strong baseline [3, 37], and episodic training on top of it with a cosine nearest-centroid head can add a little more [4]. Our stages S0 and M follow that recipe. The cross-domain benchmark of Guo et al. [13] has crop-disease and EuroSAT targets and reports that meta-learning methods often lose to simple fine-tuning when the target is far from the source. Fine-tuning a strong pre-trained backbone is a strong baseline for few-shot classification [7, 18]; those studies do not prune the backbone. Section 5 asks whether the episodic stage still helps once the network is pruned.

Earth observation. Few-shot scene classification in remote sensing has been approached with metric learning on aerial benchmarks [21] and with meta-learning for land-cover classification [30]. NWPU-RESISC45 [5] and the Sentinel-2 [8] patches of EuroSAT [16] are standard scene and land-use benchmarks, which we split by class instead of by image. Pretraining on satellite imagery, for example with masked autoencoders [6], is the natural alternative to an ImageNet initialisation; we did not test it. CloudScout [11] runs a compact CNN on a Myriad 2 accelerator in the Φ-Sat-1 mission [12]. We do not target that hardware.

Plants. PlantVillage [19] and PlantDoc [32] are common plant-disease datasets; Argüeso et al. [1] study few-shot recognition on PlantVillage with a 32/6 class split, and we use a similar cross-class split.

3. Method

Notation. S0 is cross-entropy pre-training of the ImageNet-initialised network on the base classes. R is the recovery training, with the same loss, that follows a pruning step. M is episodic (prototypical) meta-training. G is the same episodes with the learning rate set to zero, a probe that only collects gradient statistics. P is a pruning step. A mid channel is an output channel of the first convolution of a residual block, the inner channel we remove. All methods of one class partition start from the same S0 network.

Episodes and classifier. An episode draws N=5 classes with K labelled support images and Q=15 query images each. We embed an image with the global-average-pooled output of the backbone (512-d for ResNet-18), L2-normalise it, average the support embeddings of each class into a prototype, and give each query the label of the most similar prototype under cosine similarity, as in prototypical networks [33] with a cosine head [4]. Base classes are the only classes the network sees, whether for training, for pruning decisions or for the calibration statistics below. Novel classes appear at evaluation and nowhere else.

What is pruned. A ResNet-18 has eight basic blocks with two 3×3 convolutions each. We remove output channels of the first convolution, with their BatchNorm entries and the matching input slices of the second (1920 mid channels in total). The stem, the shortcuts and the block outputs keep their width, so no index has to be matched across a residual addition; He et al. [15] prune residual branches the same way. The budget is the share of the 11.18 M trunk parameters that remain, counted on the pruned network (the classifier is a parameter-free prototype head). To spread a budget over depth every score uses one rule: block $b \in \{0, \dots, 7\}$ keeps a fraction $\kappa\,(1 - \tau\, b/7)$ of its mid channels, with τ=0.5 and κ found by bisection. Deep blocks hold most of the weights, so they lose most of the channels.

Pruning MobileNetV2. To see whether the answer depends on the architecture we apply the same procedure to the torchvision MobileNetV2. Here the prunable unit is a hidden channel of an inverted-residual block (the expansion convolution, its BatchNorm, the matching depthwise filter and BatchNorm, and the input slice of the projection are cut together; 16 blocks with 7104 hidden channels). The mid activation is the output of the depthwise stage; the filter-based scores use the expansion filter, and the BatchNorm scale is that of the depthwise BatchNorm. The depth rule and the bisection are the same, the budgets are 60, 40 and 25% of the 2.22 M parameters, and only schedule P is used, because the meta stage did not help on the ResNet.

Scores. All scores are computed on a calibration set of 40 images per base class, without augmentation. Four are standard: random; the ℓ_1 norm of the filter [20]; the BatchNorm scale $|\gamma|$ of the pretrained network, which network slimming [23] ranks channels by, here without its sparsity penalty; and a first-order Taylor term [27], the mean over images of $|\sum_{x,y} a\, \partial L/\partial a|$ for the post-ReLU mid activation a. The loss L is the cross-entropy of a cosine nearest-prototype classifier whose prototypes are the calibration class means under the current network (temperature 10), so no trained head has to survive the earlier stages.

DACIS combines three per-channel quantities. $\mathcal{G}_c$ is the mean ℓ_2 norm of $\partial L/\partial W_c$ over calibration batches of 64 images, where W_c is the filter of channel c. $\mathcal{V}_c$ is the variance, across calibration images, of the spatially pooled activation of the channel. $\mathcal{D}_c$ is the Fisher ratio of that pooled activation over the class labels, $\mathcal{D}_c = \sum_k n_k(\mu_{kc} - \mu_c)^2 / \sum_k \sum_{i \in k}(a_{ic} - \mu_{kc})^2$, the between-class over within-class scatter of linear discriminant analysis [10]. Each quantity is replaced by its rank inside the block, scaled to $[0, 1]$ (written rk), and the score is

$$\mathrm{DACIS}_c = \lambda_G\, \mathrm{rk}(\mathcal{G}_c) + \lambda_V\, \mathrm{rk}(\mathcal{V}_c) + \lambda_D\, \mathrm{rk}(\mathcal{D}_c), \quad (1)$$

with weights $(\lambda_G, \lambda_V, \lambda_D) = (0.3, 0.2, 0.5)$ that we fixed before running any experiment reported here. Sec. 5 looks at them on held-out validation classes. Ranks rather than raw values keep the heavy tails of $\mathcal{D}$ and $\mathcal{G}$ from deciding the sum.

The PMP schedule. After S0, P1 prunes to 60% of the parameters, R recovers, and M runs 1000 episodes (5-way, 5-shot, 15 queries) of prototypical training with frozen BatchNorm statistics, in the manner of Meta-Baseline [4]. While M runs we accumulate, for every mid channel, the mean norm of the episode-loss gradient with respect to its filter, G_{meta}. P2 prunes to the target budget with $\mathrm{DACIS}_c\,\big(1 + \gamma\, \mathrm{rk}(G_{\mathrm{meta},c})\big)$, γ=1, recomputed on the meta-trained network (only DACIS uses G_{meta}), and a last R repairs the damage. We compare seven schedules: P, P→M, M→P, P→G→P, PMP, PMPM and P→M with twice the episodes (the supplement lists their steps). Two of them are controls. P→M with twice the episodes tests whether extra stages help only because they train longer. P→G→P replaces M by 200 episodes at learning rate zero, so the network is untouched and only the gradient statistic is

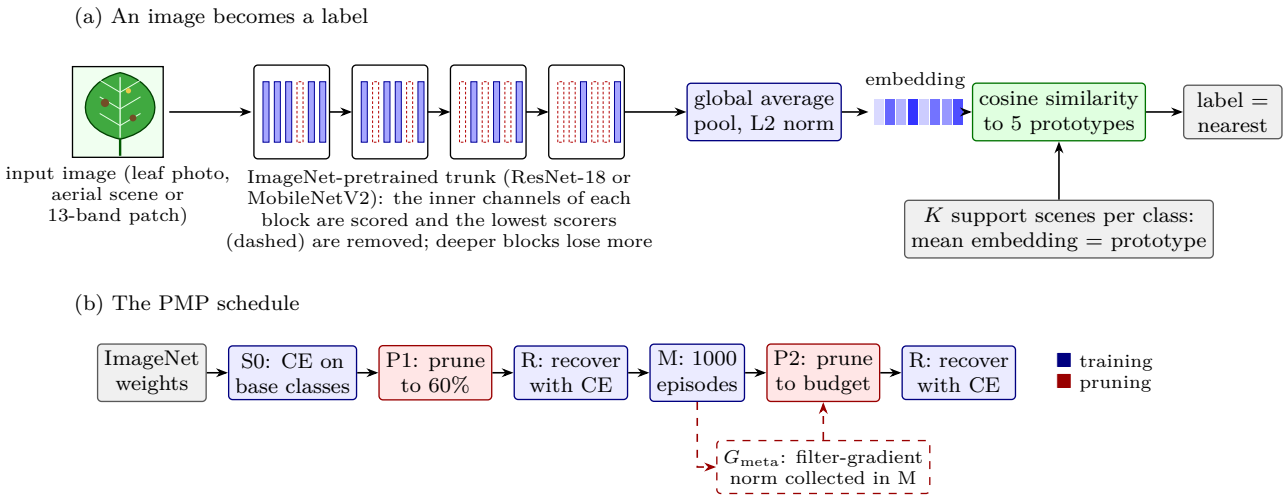


Figure 1. An image becomes a label through a pruned trunk and a nearest-prototype head (a); the PMP schedule (b).

collected; it asks whether the information in the meta stage is useful without its weight updates. The learning rate of M was chosen on validation classes (Sec. 4).

4. Experimental protocol

Data. NWPU-RESISC45 [5] has 31,500 aerial RGB scenes of 256×256 pixels, 700 for each of 45 scene classes from airport to wetland; we resize them to 128×128. EuroSAT-MS [16] has 27,000 Sentinel-2 [8] patches of 64×64 pixels (a 640 m footprint at 10 m resolution) with 13 spectral bands and 10 land-use classes. Each band is standardised with the mean and standard deviation of at most 3000 randomly chosen base-class images. To accept 13 channels the first convolution repeats the ImageNet filters over the bands and scales them by 3/13; a control network sees only the visible bands B4, B3 and B2 through the unmodified filters. As a ground-level control we use PlantVillage [19] (54,305 photographs of single leaves on a plain background, 38 classes, resized to 128×128) and PlantDoc [32], internet photographs with cluttered scenes. For PlantDoc we use the cropped-leaf release (2569 images, 28 label names); classes with fewer than 25 images (one label, with two) are dropped, which leaves 27 classes. No PlantDoc image is used for training or pruning.

Partitions and episodes. A seed fixes a random split of the classes. RESISC45 gives 25 base, 5 validation and 15 novel classes, EuroSAT-MS 5 base and 5 novel, PlantVillage 24 base, 6 validation and 8 novel. The main run used seeds 0 to 4 on every dataset. A second run on the same GPU type, with the same code and settings, added seeds 5 to 9 on RESISC45 and EuroSAT-MS for the dense and compact networks and for the ResNet-18 runs at 30% of the parameters (eight scores under PMP, five under P, and the schedule controls), and ran the MobileNetV2 pruning on all ten seeds of these two sets. Cells that rest on ten partitions therefore sit beside cells that rest on five; RESISC45 and EuroSAT-MS columns use ten partitions (Tables 1 and 3), PlantVillage and PlantDoc five, and Table 2 partitions 0 to 4 throughout. Every method of one partition sees the same split, the same S0 network and the same test episodes: 2000 episodes of 5 classes, 15 queries per class and $K \in \{1, 5, 10\}$ supports, drawn once per dataset, seed and K. PlantDoc is an evaluation set only; its 27 classes are used for 5-way episodes after training on PlantVillage base classes, so the test crosses a domain boundary. PlantDoc and PlantVillage share some concepts (tomato early blight, for one), so this is a field-image test and not a strict novel-class test.

Training. Stage S0 attaches a new linear head over the base classes and trains for 10 epochs (RESISC45), 8 (EuroSAT-MS) or 6 (PlantVillage) with SGD (Nesterov momentum 0.9, learning rate 0.02, weight decay $5{\cdot}10^{-4}$ on weight tensors, 3% linear warm-up then cosine decay, batch 128, label smoothing 0.1). Stage R reuses that head for 2 epochs at 0.01 on the two Earth-observation sets and for one epoch on PlantVillage; at test time the head is dropped and a parameter-free cosine prototype classifier is used. Stage M keeps BatchNorm statistics frozen, trains the cosine temperature (initialised to 10) and the other weights on augmented batches, and runs 1000 episodes (800 on EuroSAT-MS), two per step, with the optimiser, the

3% warm-up and cosine schedule and the weight decay of S0. Its learning rate was picked on the validation classes of the first PlantVillage partition from $\{10^{-4}, 2.5{\cdot}10^{-4}, 5{\cdot}10^{-4}, 10^{-3}, 2{\cdot}10^{-3}\}$; it chose 0.0001, the smallest value of the grid, and every dataset uses that value. We then ran the same sweep on the RESISC45 validation classes and a second one with smaller rates (10^{-5} to 10^{-4}); neither was used for selection. Training images get a random crop (scale 0.6 to 1), flips and, on PlantVillage and RESISC45, brightness and contrast jitter; mixed precision is on. Apart from that one learning rate we tuned nothing, and nothing per method. The off-the-shelf networks (MobileNetV3-Small [17], MobileNetV2 [31] and EfficientNet-B0 [35], torchvision weights trained on ImageNet [29], feature trunk only) get the S0 recipe followed by R, without any tuning of their own.

Cost. Parameters and multiply-accumulates (MACs) are counted exactly on the pruned network (11.18 M parameters for the dense ResNet-18 trunk, 11.21 M with thirteen bands, where the budgets are relative to that figure). Latency is measured in fp32 with one thread and batch size one for 128×128 RGB input, as the median of 60 runs after warm-up, on a 20-thread server CPU (model not reported by the host) for the main run and on AMD Ryzen 7 7435HS for a second measurement (nine interleaved rounds of 100 passes, pinned to one core); GPU throughput is measured at batch 128 on NVIDIA RTX PRO 6000 Blackwell Server Edition. Table entries are means over the runs of a row.

Reporting. Tables give the mean over partitions of the per-partition mean accuracy, with the sample standard deviation over partitions as subscript. That spread contains the effect of the partition and of training noise. It is much larger than the sampling error of 2000 episodes (at most 0.2 points), so we quote it whenever a difference is small. For a pair of methods we give the mean paired difference in points with a 95% t-interval over the per-partition differences; the interval assumes roughly normal differences, which five to ten partitions cannot test. In Table 1, ↑ and ↓ mark a paired difference to random choice whose 95% interval lies above or below zero, and a dash a score that was not run; the PlantDoc columns test the PlantVillage-trained networks on field photographs. Partitions, not episodes, are the replicates: episodes from one trained network are not independent draws of the training run.

5. Results

Pruning costs accuracy. At 30% of the parameters the ResNet-18 with the DACIS score under PMP loses 6.2 points on RESISC45 [+5.5,+6.9], 5.4 on EuroSAT-MS, 4.9 on PlantVillage and 8.0 on PlantDoc (Table 1). Accuracy varies far more between partitions than between methods: single EuroSAT-MS partitions of the dense network range from 54.0% to 71.2%.

Scores on the ResNet-18. Fig. 2 shows the paired differences to random choice. Under PMP the best and the worst of the eight scores are 2.5 points apart on RESISC45, 2.9 on EuroSAT-MS, 2.0 on PlantVillage and 3.0 on PlantDoc. Of the 28 paired comparisons of a computed score with random choice (seven scores, four settings), 2 favour the computed score (Taylor on RESISC45; Taylor on PlantDoc) and 2 favour random choice (BN scale on RESISC45; Fisher on RESISC45); about 1.4 would exclude zero by chance. Only the Taylor lead on RESISC45 survives a Bonferroni correction (adjusted $p = 0.005$ over these comparisons, 0.001 over the 50 of Table 1): 1.4 points [+0.9,+1.9] under PMP and 2.3 under P. DACIS is indistinguishable from random choice in all eight setting and schedule pairs (0.0 points [−0.8,+0.7] on RESISC45; pooled over the two Earth-observation sets, +0.4 points [−0.6,+1.5] under PMP) and 1.4 points behind Taylor on RESISC45 [−1.9,−0.9]. Kendall's W across partitions (unrelated rankings give $1/m$) is 0.67 on RESISC45 ($m = 10$, $p < 0.001$), where Taylor is usually first and BatchNorm scale and Fisher last, 0.27 on EuroSAT-MS ($m = 10$, $p = 0.009$) and 0.24 on PlantVillage ($m = 5$, $p = 0.304$).

Episodic training did not raise 5-shot accuracy. Adding 1000 episodes (800 on EuroSAT-MS) of prototypical training to the dense network changes 5-shot accuracy by −0.3 points on RESISC45 [−0.4,−0.1] and −0.8 on EuroSAT-MS [−1.5,0.0], and gains +0.6 points [+0.2,+1.0] at 1 shot on RESISC45. The validation sweeps (supplement), down to a rate of 10^{-5}, find no learning rate that beats stopping after S0 at 5 shots, the criterion that chose the rate; at 1 shot every rate of the RESISC45 sweep gains 0.8 to 1.9 points. Among the pruning schedules (supplement) PMP is ahead of P by +0.9 points on RESISC45 [+0.4,+1.5] and by +1.4 on EuroSAT-MS [−0.3,+3.2]. M→P is ahead by +0.9 on RESISC45 and the probe P→G→P, which updates no weight in the meta stage, by +1.2, whereas P→M is level with P (0.0). The schedules that gain are exactly those whose pruning score uses the meta-gradient term; M→P has one pruning step and P→G→P no weight update, so neither is needed. The gain lifts DACIS only to the level of random choice. Guo et al. [13] likewise found meta-learning behind fine-tuning on crop-disease and EuroSAT targets; our conclusion holds for

Score at 30% of the parameters	RESISC45 P	RESISC45 PMP	EuroSAT-MS P	EuroSAT-MS PMP	PlantVillage P	PlantVillage PMP	PlantDoc P	PlantDoc PMP
Dense ResNet-18, S0 only	$87.2_{\pm 2.5}$		$64.7_{\pm 4.9}$		$91.4_{\pm 5.6}$		$55.4_{\pm 1.3}$	
Dense ResNet-18, S0+R+M+R	$87.0_{\pm 2.5}$		$64.2_{\pm 4.7}$		$91.3_{\pm 5.6}$		$55.5_{\pm 1.1}$	
Random	$80.1_{\pm 3.6}$	$81.1_{\pm 3.0}$	$57.8_{\pm 6.7}$	$58.4_{\pm 4.6}$	$85.6_{\pm 8.1}$	$86.8_{\pm 7.7}$	$46.6_{\pm 1.3}$	$46.7_{\pm 1.7}$
Filter ℓ_1	$81.0_{\pm 3.1}^{\uparrow}$	$81.5_{\pm 3.0}$	$57.9_{\pm 6.2}$	$59.4_{\pm 5.7}$	$85.8_{\pm 8.3}$	$86.4_{\pm 7.7}$	$48.7_{\pm 0.9}^{\uparrow}$	$48.1_{\pm 0.8}$
BN scale $\lvert\gamma\rvert$	–	$80.0_{\pm 3.1}^{\downarrow}$	–	$56.8_{\pm 4.9}$	$85.1_{\pm 8.9}$	$85.1_{\pm 8.7}$	$44.4_{\pm 1.7}^{\downarrow}$	$45.5_{\pm 1.0}$
Taylor	$82.5_{\pm 2.9}^{\uparrow}$	$82.4_{\pm 3.0}^{\uparrow}$	$60.2_{\pm 6.0}^{\uparrow}$	$59.7_{\pm 6.3}$	$88.3_{\pm 7.1}^{\uparrow}$	$87.1_{\pm 7.1}$	$47.9_{\pm 1.2}$	$48.5_{\pm 1.0}^{\uparrow}$
Gradient $\mathcal{G}$	–	$81.4_{\pm 3.3}$	–	$58.9_{\pm 5.6}$	$87.3_{\pm 7.2}$	$86.7_{\pm 7.9}$	$47.6_{\pm 1.5}$	$46.9_{\pm 2.5}$
Variance $\mathcal{V}$	–	$81.5_{\pm 3.0}$	–	$57.3_{\pm 4.4}$	$86.5_{\pm 6.9}$	$86.5_{\pm 7.3}$	$47.2_{\pm 1.4}$	$47.3_{\pm 1.3}$
Fisher $\mathcal{D}$	$79.6_{\pm 3.1}$	$79.9_{\pm 2.9}^{\downarrow}$	$58.1_{\pm 6.5}$	$58.4_{\pm 6.0}$	$85.2_{\pm 8.9}$	$85.3_{\pm 8.0}$	$46.8_{\pm 0.8}$	$46.4_{\pm 1.3}$
DACIS	$80.1_{\pm 3.0}$	$81.0_{\pm 3.3}$	$57.8_{\pm 5.3}$	$59.3_{\pm 4.9}$	$86.4_{\pm 7.7}$	$86.6_{\pm 7.5}$	$47.3_{\pm 1.4}$	$47.5_{\pm 1.6}$

Table 1. Eight scores on a ResNet-18 at 30% of the parameters: 5-shot accuracy (%), mean$_{\pm \text{sd}}$ over partitions.

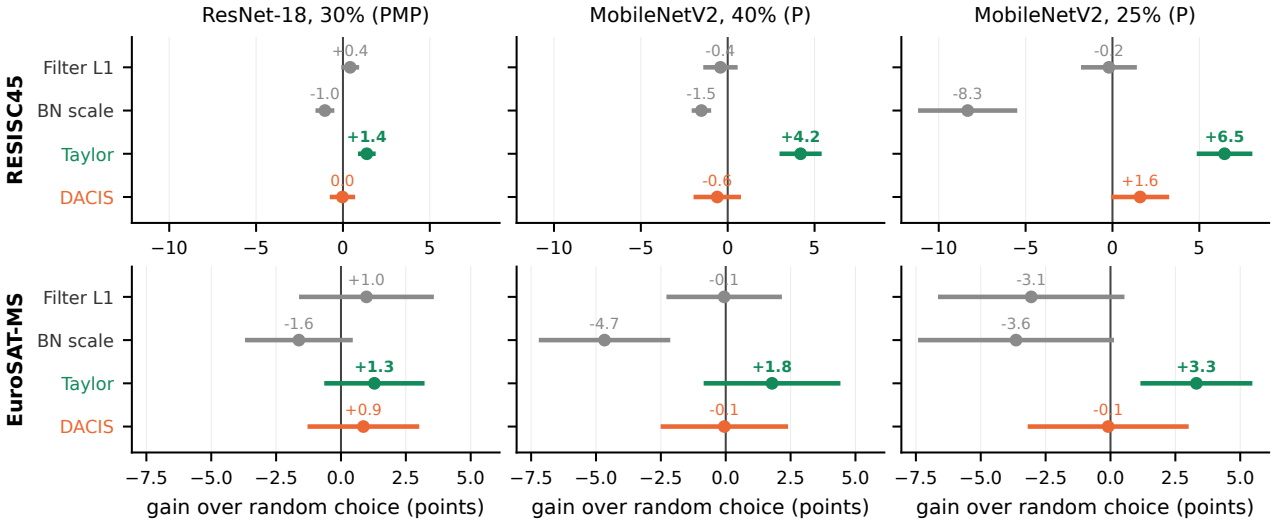


Figure 2. Gain over random choice in 5-shot points, 95% intervals over partitions: only Taylor is clearly above zero.

our recipe only (frozen BatchNorm, 1000 episodes, 5-way 5-shot).

Budget, time and the compact networks. Accuracy falls steadily with the budget (partitions 0 to 4): on RESISC45 DACIS drops from 83.8% at 50% of the parameters to 71.9% at 10%, against a spread of 2.5 points among scores at 30%. Pruning saves less time than operations (Table 2; CPU is single-thread latency at batch size 1, GPU throughput is at batch size 128, accuracy is for DACIS under PMP on the ResNet-18 rows and for Taylor under PMP or P where marked): at 30% the ResNet-18 needs 0.251 G multiply-accumulates against 0.592 G for the dense network, 58% fewer, but its single-thread CPU latency is only 39% lower (39% on a laptop CPU, AMD Ryzen 7 7435HS). We did not profile layer by layer; the untouched stem, shortcuts and residual widths, or memory traffic and call overhead at batch size one, could explain the gap. Unpruned compact networks beat every pruned ResNet-18 on the novel-class sets (Fig. 3). Over ten partitions MobileNetV3-Small, with 0.93 M parameters and 0.018 G multiply-accumulates, is 7.1 points ahead of the 30% ResNet-18 on RESISC45 and 15.4 on EuroSAT-MS (DACIS under PMP; against Taylor, 5.7 and 15.0), and runs in 8.8 ms on the laptop CPU against 40.6 ms for the dense ResNet-18. EfficientNet-B0 is higher still on RESISC45 and PlantVillage. PlantDoc is the exception: a ResNet-18 kept at 50% is 1.6 points ahead of MobileNetV3-Small there.

The score matters more under severe pruning. We repeated the comparison on MobileNetV2 (Table 3). At

Model	Params (M)	MACs (G)	CPU ms	GPU img/s	RESISC45	EuroSAT-MS	PlantVillage
ResNet-18, dense	11.18	0.592	10.8	36644	$86.9_{\pm 3.4}$	$62.1_{\pm 5.3}$	$91.4_{\pm 5.6}$
ResNet-18, 50% (DACIS)	5.59	0.398	8.6	38034	$83.8_{\pm 3.7}$	$58.6_{\pm 5.7}$	$88.6_{\pm 7.2}$
ResNet-18, 30% (DACIS)	3.35	0.251	6.6	40434	$80.5_{\pm 4.2}$	$57.1_{\pm 4.9}$	$86.6_{\pm 7.5}$
ResNet-18, 20% (DACIS)	2.24	0.178	5.7	41645	$77.4_{\pm 4.9}$	$54.5_{\pm 8.1}$	$83.8_{\pm 8.2}$
ResNet-18, 10% (DACIS)	1.11	0.105	4.6	45865	$71.9_{\pm 4.8}$	$51.4_{\pm 7.5}$	$78.2_{\pm 10.2}$
ResNet-18, 30% (Taylor)	3.35	0.251	6.6	40434	$82.2_{\pm 3.9}$	$57.0_{\pm 8.0}$	$87.1_{\pm 7.1}$
MobileNetV2, dense	2.22	0.098	4.3	24400	$86.8_{\pm 2.9}$	$64.7_{\pm 4.7}$	–
MobileNetV2, 60% (Taylor)	1.33	0.068	3.9	32373	$85.3_{\pm 3.3}$	$62.8_{\pm 4.1}$	–
MobileNetV2, 40% (Taylor)	0.89	0.041	3.0	52047	$82.6_{\pm 3.5}$	$61.2_{\pm 6.8}$	–
MobileNetV2, 25% (Taylor)	0.56	0.022	2.5	63624	$73.5_{\pm 5.8}$	$55.3_{\pm 7.2}$	–
MobileNetV3-Small, dense	0.93	0.018	2.2	63526	$88.1_{\pm 3.3}$	$72.4_{\pm 4.9}$	$91.1_{\pm 5.9}$
EfficientNet-B0, dense	4.01	0.126	6.3	18946	$90.5_{\pm 2.4}$	$69.9_{\pm 5.1}$	$93.8_{\pm 5.2}$

Table 2. Cost and 5-shot accuracy (%) of dense, pruned and compact networks (partitions 0 to 4).

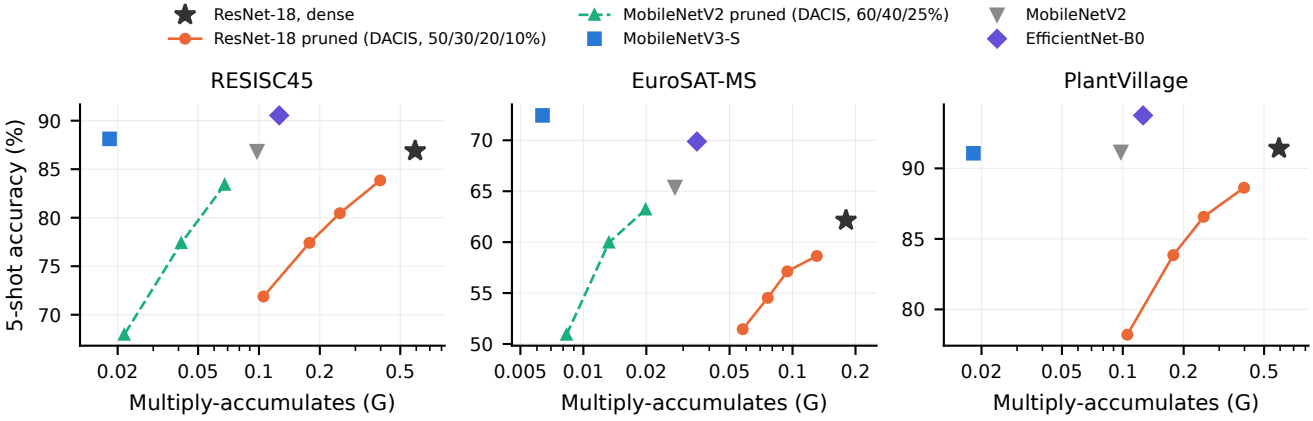


Figure 3. 5-shot accuracy against multiply-accumulates (log scale) for pruned and unpruned networks.

40% of the parameters Taylor is 4.2 points above random choice on RESISC45 [+3.0,+5.4], in 10 of 10 partitions, and 6.5 points at 25%, where BatchNorm scale falls 8.3 points below it. The Taylor lead grows with severity on both networks: on RESISC45 it is +0.8, +4.2 and +6.5 points at 60, 40 and 25% of the MobileNetV2 and +0.6, +1.4, +2.5 and +6.0 at 50, 30, 20 and 10% of the ResNet-18 (five partitions except at 30%), so these runs do not separate architecture from severity. 5 of the 24 MobileNetV2 comparisons with random choice survive a Bonferroni correction, all on RESISC45; on EuroSAT-MS Taylor is ahead by 1.8 and 3.3 points at 40 and 25%, but neither survives. DACIS is ahead of random choice in 0 of its 6 comparisons and behind it in 1 (60% on RESISC45, −1.1 points). Pruned to 40% with the Taylor term, a MobileNetV2 reaches 83.0% on RESISC45 with 0.041 G multiply-accumulates, 0.6 points above the Taylor-pruned ResNet-18, which needs 6 times as many; the unpruned MobileNetV3-Small is still ahead of it by 5.0 points.

Further checks. On the 5 EuroSAT-MS partitions that have both inputs, the dense network scores 62.1% at 5 shots with thirteen bands and 68.7% with the three visible ones, a difference of −6.6 points [−17.4,+4.2]; thirteen bands are ahead in 2 of the 5 partitions. Our first convolution copies the ImageNet RGB filters over all bands, so this tests that adaptation and not the bands themselves. Of the 9 variants of DACIS in the supplement, only dropping the meta-gradient term has an interval clear of zero on PlantVillage (−0.9 points [−1.7,−0.1]), which one variant in nine does not make surprising. On the PlantVillage validation classes, the eight weightings of Eq. (1) that we tried span 87.9% to 88.9% at 5 shots against 88.5% for the weights fixed in advance, so we left λ alone.

	RESISC45			EuroSAT-MS		
Parameters kept	60%	40%	25%	60%	40%	25%
MobileNetV2, dense		$87.4_{\pm 2.5}$			$67.4_{\pm 7.6}$	
Random	$85.1_{\pm 2.8}$	$78.8_{\pm 3.9}$	$67.0_{\pm 4.5}$	$67.3_{\pm 6.5}$	$62.1_{\pm 7.5}$	$55.3_{\pm 8.3}$
Filter ℓ_1	$84.6_{\pm 2.7}$	$78.4_{\pm 3.4}$	$66.8_{\pm 4.6}$	$66.3_{\pm 7.9}$	$62.1_{\pm 8.2}$	$52.3_{\pm 9.8}$
BN scale $\lvert\gamma\rvert$	$85.5_{\pm 2.7}$	$77.3_{\pm 3.7}$	$58.7_{\pm 5.5}$	$63.7_{\pm 8.4}$	$57.4_{\pm 9.5}$	$51.7_{\pm 10.6}$
Taylor	$85.9_{\pm 2.5}$	$83.0_{\pm 2.8}$	$73.4_{\pm 4.2}$	$66.2_{\pm 7.6}$	$63.9_{\pm 7.1}$	$58.6_{\pm 7.6}$
DACIS	$84.0_{\pm 2.7}$	$78.2_{\pm 3.1}$	$68.6_{\pm 2.8}$	$65.4_{\pm 8.0}$	$62.1_{\pm 6.0}$	$55.2_{\pm 8.5}$

Table 3. MobileNetV2 pruned with five scores (schedule P): 5-shot accuracy (%), mean over ten partitions.

6. Limitations

Five to ten class partitions are still few. The paired intervals have a mean half-width of 0.6 points on RESISC45 and 2.1, 2.1 and 1.5 on EuroSAT-MS, PlantVillage and PlantDoc, so "indistinguishable" means "not distinguishable at that width", not "equal". Each partition has one training run per method, which mixes training noise with the method effect, and the 28 comparisons share one random baseline and the same partitions. Rerunning the same code and seeds on an NVIDIA GeForce RTX 3050 A Laptop GPU changed the 5-shot accuracy of a single run by 1.0 points on average and by up to 5.1 points over 95 runs of 2 partitions. The episodic stage was tested with one recipe, and one learning rate, chosen on PlantVillage at the smallest value of its grid, was used everywhere. The compact networks were not tuned.

"Novel" means novel with respect to the base classes, not to ImageNet, which has seen airplanes, ships and tennis courts. EuroSAT-MS has ten classes, so its novel-class numbers rest on a five-class base set, and the comparison of thirteen bands with three depends on how the first convolution was initialised. Nothing here uses coordinates, acquisition dates or radar, and nothing was run on full scenes, so what is specific to geospatial data is absent. RESISC45 scenes were resized from 256 to 128 pixels, which changes the ground sampling distance, and the class-disjoint split does not separate neighbouring tiles geographically. PlantVillage holds several photographs of the same leaf that we did not keep apart between support and query, so its numbers may be high.

Cost is parameters, multiply-accumulates, single-thread CPU latency and GPU throughput; we ran nothing on a Raspberry Pi, a Jetson or an on-board accelerator and measured no energy. Only the ResNet-18 and the MobileNetV2 were pruned, and only their inner channels.

7. Conclusion

For a ResNet-18 asked to recognise unseen aerial and Sentinel-2 classes from a few examples, which channels were removed mattered little at 30% of the parameters: eight scores, random choice among them, land 2.0 to 3.0 points apart on four test settings. DACIS is never reliably ahead of random choice. Only the Taylor term is, by 1.4 points on RESISC45, and its lead grows with the severity of pruning (to 6.0 points at 10% of the ResNet-18 and 6.5 at 25% of a MobileNetV2), where BatchNorm scale does worse than random choice. The meta stage did not raise 5-shot accuracy. Pruning saved less time (39%) than operations (58%), and an unpruned MobileNetV3-Small beat every pruned network we built on the three novel-class sets.

Before pruning anything for few-shot classification of optical patches we would test a small ImageNet-initialised network such as MobileNetV3-Small. If pruning is needed, we would use the Taylor term, skip the meta stage, and report many class partitions with paired episodes. Satellite-pretrained backbones and embedded hardware remain open.

References

[1] David Argüeso, Artzai Picon, Unai Irusta, Alfonso Medela, Miguel G. San-Emeterio, Arantza Bereciartua, and Aitor Alvarez-Gila. Few-shot learning approach for plant disease classification using images taken in the field. Comput. Electron. Agric., 175: 105542, 2020. 2

[2] Davis Blalock, Jose Javier Gonzalez Ortiz, Jonathan Frankle, and John Guttag. What is the state of neural network pruning? In Proc. Mach. Learn. Syst., 2020. 2

[3] Wei-Yu Chen, Yen-Cheng Liu, Zsolt Kira, Yu-Chiang Frank Wang, and Jia-Bin Huang. A closer look at few-shot classification. In Int. Conf. Learn. Represent., 2019. 2

[4] Yinbo Chen, Zhuang Liu, Huijuan Xu, Trevor Darrell, and Xiaolong Wang. Meta-baseline: Exploring sim-

ple meta-learning for few-shot learning. In Int. Conf. Comput. Vis., pages 9062–9071, 2021. 2, 3

[5] Gong Cheng, Junwei Han, and Xiaoqiang Lu. Remote sensing image scene classification: Benchmark and state of the art. Proc. IEEE, 105(10):1865–1883, 2017. 1, 2, 4

[6] Yezhen Cong, Samar Khanna, Chenlin Meng, Patrick Liu, Erik Rozi, Yutong He, Marshall Burke, David B. Lobell, and Stefano Ermon. SatMAE: Pre-training transformers for temporal and multi-spectral satellite imagery. In Adv. Neural Inform. Process. Syst., 2022. 2

[7] Guneet S. Dhillon, Pratik Chaudhari, Avinash Ravichandran, and Stefano Soatto. A baseline for few-shot image classification. In Int. Conf. Learn. Represent., 2020. 2

[8] M. Drusch, U. Del Bello, S. Carlier, O. Colin, V. Fernandez, F. Gascon, B. Hoersch, C. Isola, P. Laberinti, P. Martimort, A. Meygret, F. Spoto, O. Sy, F. Marchese, and P. Bargellini. Sentinel-2: ESA's optical high-resolution mission for GMES operational services. Remote Sens. Environ., 120:25–36, 2012. 2, 4

[9] Chelsea Finn, Pieter Abbeel, and Sergey Levine. Model-agnostic meta-learning for fast adaptation of deep networks. In Int. Conf. Mach. Learn., 2017. 2

[10] Ronald A. Fisher. The use of multiple measurements in taxonomic problems. Ann. Eugen., 7(2):179–188, 1936. 3

[11] Gianluca Giuffrida, Lorenzo Diana, Francesco de Gioia, Gionata Benelli, Gabriele Meoni, Massimiliano Donati, and Luca Fanucci. CloudScout: A deep neural network for on-board cloud detection on hyperspectral images. Remote Sens., 12(14):2205, 2020. 1, 2

[12] Gianluca Giuffrida, Luca Fanucci, Gabriele Meoni, et al. The ϕ-sat-1 mission: The first on-board deep neural network demonstrator for satellite earth observation. IEEE Trans. Geosci. Remote Sens., 60:1–14, 2022. 1, 2

[13] Yunhui Guo, Noel C. Codella, Leonid Karlinsky, James V. Codella, John R. Smith, Kate Saenko, Tajana Rosing, and Rogerio Feris. A broader study of cross-domain few-shot learning. In Eur. Conf. Comput. Vis., 2020. 2, 5

[14] Kaiming He, Xiangyu Zhang, Shaoqing Ren, and Jian Sun. Deep residual learning for image recognition. In IEEE Conf. Comput. Vis. Pattern Recog., 2016. 1

[15] Yihui He, Xiangyu Zhang, and Jian Sun. Channel pruning for accelerating very deep neural networks. In Int. Conf. Comput. Vis., 2017. 1, 2, 3

[16] Patrick Helber, Benjamin Bischke, Andreas Dengel, and Damian Borth. EuroSAT: A novel dataset and deep learning benchmark for land use and land cover classification. IEEE J. Sel. Top. Appl. Earth Obs. Remote Sens., 12(7):2217–2226, 2019. 1, 2, 4

[17] Andrew Howard, Mark Sandler, Grace Chu, Liang-Chieh Chen, Bo Chen, Mingxing Tan, Weijun Wang, Yukun Zhu, Ruoming Pang, Vijay Vasudevan, Quoc V. Le, and Hartwig Adam. Searching for MobileNetV3. In Int. Conf. Comput. Vis., 2019. 5

[18] Shell Xu Hu, Da Li, Jan Stühmer, Minyoung Kim, and Timothy M. Hospedales. Pushing the limits of simple pipelines for few-shot learning: External data and fine-tuning make a difference. In IEEE/CVF Conf. Comput. Vis. Pattern Recognit., 2022. 2

[19] David P. Hughes and Marcel Salathé. An open access repository of images on plant health to enable the development of mobile disease diagnostics. arXiv preprint arXiv:1511.08060, 2015. 1, 2, 4

[20] Hao Li, Asim Kadav, Igor Durdanovic, Hanan Samet, and Hans Peter Graf. Pruning filters for efficient convnets. In Int. Conf. Learn. Represent., 2017. 1, 2, 3

[21] Haifeng Li, Zhenqi Cui, Zhiqing Zhu, Li Chen, Jiawei Zhu, Haozhe Huang, and Chao Tao. RS-MetaNet: Deep meta metric learning for few-shot remote sensing scene classification. IEEE Trans. Geosci. Remote Sens., 59(8):6983–6994, 2021. 2

[22] Yawei Li, Kamil Adamczewski, Wen Li, Shuhang Gu, Radu Timofte, and Luc Van Gool. Revisiting random channel pruning for neural network compression. In IEEE/CVF Conf. Comput. Vis. Pattern Recognit., 2022. 2

[23] Zhuang Liu, Jianguo Li, Zhiqiang Shen, Gao Huang, Shoumeng Yan, and Changshui Zhang. Learning efficient convolutional networks through network slimming. In Int. Conf. Comput. Vis., 2017. 1, 2, 3

[24] Zechun Liu, Haoyuan Mu, Xiangyu Zhang, Zichao Guo, Xin Yang, Kwang-Ting Cheng, and Jian Sun. MetaPruning: Meta learning for automatic neural network channel pruning. In Int. Conf. Comput. Vis., pages 3296–3305, 2019. 2

[25] Zhuang Liu, Mingjie Sun, Tinghui Zhou, Gao Huang, and Trevor Darrell. Rethinking the value of network pruning. In Int. Conf. Learn. Represent., 2019. 2

[26] Deepak Mittal, Shweta Bhardwaj, Mitesh M. Khapra, and Balaraman Ravindran. Recovering from random pruning: On the plasticity of deep convolutional neural networks. In IEEE Winter Conf. Appl. Comput. Vis., pages 848–857, 2018. 2

[27] Pavlo Molchanov, Stephen Tyree, Tero Karras, Timo Aila, and Jan Kautz. Pruning convolutional neural networks for resource efficient inference. In Int. Conf. Learn. Represent., 2017. 1, 2, 3

[28] Pavlo Molchanov, Arun Mallya, Stephen Tyree, Iuri Frosio, and Jan Kautz. Importance estimation for neural network pruning. In IEEE Conf. Comput. Vis. Pattern Recog., 2019. 2

[29] Olga Russakovsky, Jia Deng, Hao Su, Jonathan Krause, Sanjeev Satheesh, Sean Ma, Zhiheng Huang, Andrej Karpathy, Aditya Khosla, Michael Bernstein, Alexander C. Berg, and Li Fei-Fei. ImageNet large scale visual recognition challenge. Int. J. Comput. Vis., 115(3):211–252, 2015. 5

[30] Marc Rußwurm, Sherrie Wang, Marco Körner, and David Lobell. Meta-learning for few-shot land cover

classification. In IEEE/CVF Conf. Comput. Vis. Pattern Recognit. Workshops, 2020. 2

[31] Mark Sandler, Andrew Howard, Menglong Zhu, Andrey Zhmoginov, and Liang-Chieh Chen. MobileNetV2: Inverted residuals and linear bottlenecks. In IEEE/CVF Conf. Comput. Vis. Pattern Recognit., 2018. 1, 5

[32] Davinder Singh, Naman Jain, Pranjali Jain, Pratik Kayal, Sudhakar Kumawat, and Nipun Batra. PlantDoc: A dataset for visual plant disease detection. In ACM India Joint Int. Conf. Data Sci. Manage. Data (CoDS-COMAD), pages 249–253, 2020. 1, 2, 4

[33] Jake Snell, Kevin Swersky, and Richard S. Zemel. Prototypical networks for few-shot learning. In Adv. Neural Inform. Process. Syst., 2017. 2, 3

[34] Jingtong Su, Yihang Chen, Tianle Cai, Tianhao Wu, Ruiqi Gao, Liwei Wang, and Jason D. Lee. Sanity-checking pruning methods: Random tickets can win the jackpot. In Adv. Neural Inform. Process. Syst., 2020. 2

[35] Mingxing Tan and Quoc V. Le. EfficientNet: Rethinking model scaling for convolutional neural networks. In Int. Conf. Mach. Learn., 2019. 5

[36] Hongduan Tian, Bo Liu, Xiao-Tong Yuan, and Qingshan Liu. Meta-learning with network pruning. In Eur. Conf. Comput. Vis., 2020. 2

[37] Yonglong Tian, Yue Wang, Dilip Krishnan, Joshua B. Tenenbaum, and Phillip Isola. Rethinking few-shot image classification: A good embedding is all you need? In Eur. Conf. Comput. Vis., 2020. 2

[38] Zhuangwei Zhuang, Mingkui Tan, Bohan Zhuang, Jing Liu, Yong Guo, Qingyao Wu, Junzhou Huang, and Jinhui Zhu. Discrimination-aware channel pruning for deep neural networks. In Adv. Neural Inform. Process. Syst., 2018. 2